

1(a)

$$s_y = u_y t + \frac{1}{2} g t^2$$

From Fig. 1.1,  $s_y = 1.75$  m and  $t = 0.600$  s [1]

$$\begin{aligned} \therefore g &= \frac{2s_y}{t^2} = \frac{2(1.75)}{(0.600)^2} \\ &= 9.722 \text{ m s}^{-2} \\ &= 9.72 \text{ m s}^{-2} \quad [1] \end{aligned}$$

(b) % uncertainty = actual uncertainty/ data point. [1]

Hence the larger the data point, the smaller the % uncertainty since the absolute uncertainty is fixed. Hence more reliable. [1]

(c)

$$\frac{\Delta g}{g} = \frac{\Delta s_y}{s_y} + 2 \frac{\Delta t}{t} \quad [1]$$

$$\begin{aligned} \Delta g &= \left[ \frac{0.001}{1.75} + 2 \left( \frac{0.006}{0.600} \right) \right] (9.722) \\ &= 0.2 \text{ m s}^{-2} \quad [1] \end{aligned}$$

$$g = 9.7 \pm 0.2 \text{ m s}^{-2} \quad [1]$$

(d) Since  $s_y = \frac{1}{2} g t^2$ , plot a graph of  $s_y$  against  $t^2$ , where,  $s_y$  = vertical distance travelled by the sphere,  $t$  = time taken to travel  $s_y$ . The gradient =  $\frac{1}{2}g$ . [1]

Random error is reduced when a best fit line is drawn using all the data points. [1]

2(a) By the principle of conservation of momentum, since there is no external force acting, the